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oceans. They measure parameters such as temperature, salinity, current speed, gas fluxes, chemical components.

7. **Physics in relation to psychology:** All psychological interactions can be derived from a physical process. The movements of neurotransmitters are governed by the physical properties of diffusion and molecular motion. The functioning of our brain is related to our underlying dualism.

Nature teaches true science with physics as an efficient tool. Science and technology should be used in a balanced manner so that they do not become weapons to destroy nature which taught us science. Global warming and other negative impacts of technology need to be checked. Safe science with moderate and appropriate use of technology is the need of this century.

The scope and opportunities for higher education in physics and various fellowships offered is given in the beginning of the book.

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## 1.4 MEASUREMENT

*"When you can measure what you are speaking about and can express it in numbers, you know something about it; but when you cannot express it in numbers, your knowledge is of a meager and unsatisfactory kind"* - **Lord Kelvin**

The comparison of any physical quantity with its standard unit is known as measurement.

Measurement is the basis of all scientific studies and experimentation. It plays an important role in our daily life. Physics is a quantitative science and physicists always deal with numbers which are the measurement of physical quantities.

### 1.4.1 Definition of Physical Quantity

Quantities that can be measured, and in terms of which, laws of physics are described are called physical quantities. Examples are length, mass, time, force, energy, etc.

### 1.4.2 Types of Physical Quantities

Physical quantities are classified into two types. They are fundamental and derived quantities.

*Fundamental or base quantities are quantities which cannot be expressed in terms of any other physical quantities.* These are length, mass, time, electric current, temperature, luminous intensity and amount of substance.

*Quantities that can be expressed in terms of fundamental quantities are called derived quantities.* For example, area, volume, velocity, acceleration, force.